

Requirements Analysis and Specification Document (RASD)

F1tenth PoliMI

The car: base platform and advanced topic

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1 Executive Summary

1.1 Overview

We build a 1/10-scale autonomous car (F1TENTH) and its ROS 2 software stack. The car runs indoors, in a room, on a closed track drawn on the floor with tape. It maps the room with a LiDAR, finds its pose in that map, sees the tape with a camera and drives laps without leaving the lines. A person can always stop it or take it over.

The same stack runs on a simulator, so every function can be tried before the car. The professor will use the car and the stack for the summer school exercises (esercitazioni).

1.2 Project Objectives

- **Base platform (required):** a complete car and stack that works end to end: drivers, simulator, mapping, localization, line detection, path following and safety, with the documentation another team needs to use it.
- **One advanced topic:** racing-line planning (option A) or hybrid localization (option B) **[TBD: OI-01]**.
- **Key deliverable:** the 30L delivery, what our team delivers to get 30 e lode: the base platform plus one advanced topic.

1.3 Scope

In scope: the car's hardware as selected in the master BOM, the ROS 2 stack on the car and the stack's use of the simulator, the safety features, the documentation.

Out of scope here: the simulator's internals (the gym, the bridge and our `polimi_sim` extensions). Their design is in `TDD_Gym`. This document only requires that the simulator stands in for the car (section 2.3.3).

Reference, not required: the summer school exercises (section 2.5).

> **Key point.** The LiDAR maps the room, the camera sees the track. Tape is flat on the floor and the LiDAR scans a plane above it, so nothing the LiDAR sees marks the track. Every requirement below follows from that.

2 Requirements Analysis

2.1 Business Requirements

2.1.1 Business Goals

1. **30 e lode:** deliver a state-of-the-art base platform and one advanced topic.
2. **Summer school:** the professor can teach the exercises on our car and stack.
3. **Hand-over:** another team or the professor can set up, run and extend the platform without us.

2.1.2 Stakeholder Analysis

Stakeholder	Interest
Our team	Grade (30L): a platform that works end to end and one advanced topic that measurably improves on it.

Professor	A car and stack to teach the summer school exercises on; documentation to run it without us.
Summer school students	Exercises that run in the simulator and on the car, from the simplest to the most advanced.
Future teams	A platform to extend: clear interfaces, setup and design decisions written down.

2.1.3 Legend

- **Phase:** `V1` base platform; `V2` advanced topic; `Ref` reference only (summer school exercises, **[TBD: OI-12]**).
- **Src:** `S` the project scope (`docs/scope.md` at `425ee04`, folded here); `H` the hardware studies (`TDD_F1tenth`, Hardware Selection); `B` master BOM; `D` a decided open item; `C` the car bringup notes (folded into `TDD_F1tenth`).
- **IDs:** `FR-<DOM>-nnn` functional, `NFR-<CAT>-nnn` non-functional, `C-nnn` constraint, `A-nnn` assumption.

2.2 Constraints and Assumptions

2.2.1 Environment

ID	Constraint	Src
C-001	The car runs indoors, in a room.	S
C-002	The track is a closed loop drawn on the floor with tape. It has no walls or barriers: nothing along the track rises above the floor.	S
C-003	The only things standing up from the floor are the room itself: its walls, furniture and anything else in it.	S
C-004	The LiDAR cannot see the track: it scans a horizontal plane above the tape and sees the room only.	S
C-005	Leaving the track is not a collision: nothing physical stops the car when it crosses a line.	S

2.2.2 Hardware

ID	Constraint	Src
C-010	Chassis: Traxxas Ford Fiesta ST Rally VXL (74276-4), 1/10 scale, brushless, 4WD. End of life, available through unofficial retailers.	B
C-011	Compute: NVIDIA Jetson Orin Nano Super with a 500 GB NVMe SSD.	B
C-012	LiDAR: Orbbec Pulsar SL450: 270°, up to 40 Hz at 0.2°, Ethernet. It takes the Orin Nano's only Ethernet port.	B, H
C-013	Camera: Orbbec Gemini 335L: global-shutter colour 1280×800 up to 60 fps, 94°×68°, depth aligned to colour, IMU, USB 3.	B, H
C-014	Motor and servo through a VESC 6 MkVI; power through the RoboRacer power board; manual control through an 8BitDo Ultimate 2C wireless joystick.	B

ID	Constraint	Src
C-015	The track material (ducts) is not bought: the track is tape.	B, S

2.2.3 Software

ID	Constraint	Src
C-020	ROS 2 Jazzy on every image, Mac and car. Humble ends on 31 May 2027, before or during the summer school; Jazzy is supported to 31 May 2029 (OI-13, decided 21 Oct 2026).	D
C-021	The stack runs in Docker, the same on the simulator and on the car.	S
C-022	Drivers come from our fork of `f1tenth_system` (branch `jazzy-polimi`); the LiDAR and the camera are driven by Orbbec's `orbbec_camera`.	S, D
C-023	The simulator is upstream's JAX gym (`dev-jax`) and bridge (`dev-jazzy`) with our extensions in our fork of `f1tenth_gym_ros` (OI-13).	D

2.2.4 Assumptions

ID	Assumption	Src
A-001	The room is not too large, and it is mapped with SLAM on site.	S
A-002	Floor material and colour are not defined yet; ideally they vary too. **[TBD]**	S
A-003	Tape colour and width are not defined yet. **[TBD]**	S

2.3 Functional Requirements

2.3.1 User Roles

Role	Description	Responsibilities
Operator	The person next to the car, holding the joystick	Starts and stops runs, holds the deadman, takes over, presses the emergency stop
Developer	Team member or student writing stack code	Tries every change in the simulator first, then on the car
Maintainer	Our team, later another team	Builds the images, keeps the forks in sync, records design decisions

2.3.2 Use Cases

> **UC-001 — Map the room**
> **Actor:** Operator
> **Precondition:** car in the room, stack running, joystick connected
> **Main flow:** 1. Operator drives the car around the room by joystick; 2. SLAM builds the room map; 3. Operator saves the map

Postcondition: a saved map of the room (not of the track)

UC-002 — Drive laps on the track centre seen by the camera
Actor: Operator
Precondition: car on the track, stack running
Main flow: 1. Operator holds the deadman; 2. line detection publishes the track centre as a path in the car frame; 3. the path follower drives along it
Postcondition: full laps without leaving the lines; releasing the deadman stops the car

UC-003 — Drive laps on a path in the map
Actor: Operator
Precondition: saved room map, a saved path in map coordinates **[TBD: OI-03]**
Main flow: 1. Operator gives the initial pose; 2. localization tracks the car in the map; 3. the path is published once; 4. the path follower drives along it
Postcondition: full laps without leaving the lines

UC-004 — Stop or take over
Actor: Operator
Precondition: the autonomy is driving
Main flow: the Operator presses the emergency stop, releases the deadman, or drives with the joystick; or an obstacle appears in front of the car
Postcondition: the car stops or follows the joystick, whatever the autonomy sends

UC-005 — Try it in the simulator first
Actor: Developer
Precondition: the container on a Mac or PC
Main flow: 1. Developer starts the simulator as the car; 2. runs the same stack and launch as on the car; 3. then runs it on the car unchanged
Postcondition: the behaviour seen in the simulator is the behaviour on the car, within the simulator's models

2.3.3 Detailed Functional Requirements

Drivers (DRV)

ID	Requirement	Phase	Src
FR-DRV-001	The stack shall drive the motor and the servo through the VESC and publish wheel odometry from it.	V1	S
FR-DRV-002	The stack shall read the joystick and turn it into drive commands while the operator holds the enable button.	V1	S
FR-DRV-003	The stack shall publish the Pulsar SL450 scans and the Gemini 335L colour images, camera info and IMU through `orbbec_camera`.	V1	S
FR-DRV-004	The stack shall publish the mounting transforms from `base_link` to the LiDAR and to the camera, from measured values.	V1	C

Mapping (MAP)

ID	Requirement	Phase	Src
FR-MAP-001	The stack shall build a map of the room with SLAM (for example `slam_toolbox`) and save it.	V1	S
FR-MAP-002	The map shall be of the room, not of the track.	V1	S

Localization (LOC)

ID	Requirement	Phase	Src
FR-LOC-001	The stack shall localize the car in the saved room map with AMCL, from odometry and LiDAR scans. Node choice **[TBD: OI-07]**.	V1	S
FR-LOC-002	The stack shall accept an initial pose from the operator.	V1	S
FR-LOC-003	The stack should fuse wheel odometry and the camera IMU into `odom` with an EKF. **[TBD: OI-08]**	V1	S

Line detection (LIN)

ID	Requirement	Phase	Src
FR-LIN-001	The stack shall detect the tape lines in the camera image. Method **[TBD: OI-06]**.	V1	S
FR-LIN-002	The stack shall publish the track centre as a `nav_msgs/Path`, recomputed from every camera frame, covering a short stretch ahead of the car. Frame **[TBD: OI-02]**.	V1	S

Path following (PTH)

ID	Requirement	Phase	Src
FR-PTH-001	A PID controller shall drive the car on the lateral (and heading) error between the car's pose and a path. Form **[TBD: OI-05]**.	V1	S
FR-PTH-002	The path shall be a `nav_msgs/Path` on one topic, in any frame: drawn by the user (map), from line detection (car frame), or the racing line of option A (map).	V1	S
FR-PTH-003	The pose shall be read from TF in the path's frame. A path in the car frame shall need no map pose.	V1	S
FR-PTH-004	The controller shall publish `/drive`. Any node with the same inputs (path, pose) and output shall be able to replace it, for example Pure Pursuit.	V1	S
FR-PTH-005	Offline paths (hand-drawn, racing line) shall be computed once, saved to a file and published once as a closed loop (latched). Open paths **[TBD: OI-04]**.	V1	S
FR-PTH-006	On a full loop the controller shall search the closest point only in a window after the last one it used; on a fragment it shall search all of it.	V1	S

ID	Requirement	Phase	Src
FR-PTH-007	If an online path stops updating (the camera loses the tape), the controller shall stop the car. A latched offline path is never stale.	V1	S
FR-PTH-008	The stack shall provide a path drawn in the map. Source **[TBD: OI-03]**: teach and repeat suggested for V1, tape mapping later.	V1	S

Safety (SAF)

ID	Requirement	Phase	Src
FR-SAF-001	A person with the joystick shall always be able to override the autonomy: the joystick has priority over `/drive`.	V1	S
FR-SAF-002	An emergency stop shall stop the car whatever is driving. Mechanism **[TBD: OI-09]**.	V1	S
FR-SAF-003	The autonomy shall drive only while the operator holds the deadman; releasing it shall stop the car. Mechanism **[TBD: OI-09]**.	V1	S, C
FR-SAF-004	Automatic emergency braking (AEB) shall compute the time to collision from the LiDAR and brake under a threshold, whatever is driving, while still letting a person drive the car away.	V1	S
FR-SAF-005	Each command input shall time out: a source that stops sending loses control of the car.	V1	C

Simulator (SIM)

ID	Requirement	Phase	Src
FR-SIM-001	The simulator shall stand in for the car with the same topics, frames and launch, so the stack cannot tell which one it runs on.	V1	S, C
FR-SIM-002	Every safety feature (FR-SAF-001 to 004) shall work in the simulator as on the car.	V1	S
FR-SIM-003	The stack shall never read the simulator's ground truth; it is for debugging and scoring only.	V1	S

Documentation (DOC)

ID	Requirement	Phase	Src
FR-DOC-001	The platform shall be documented: setup, how to run each part, and design decisions, so another team or the professor can use it without us.	V1	S

Advanced topic (ADV)

ID	Requirement	Phase	Src
FR-ADV-001	**Option A, racing line.** The stack shall get the track boundaries in map coordinates by mapping the tape lines seen by the camera into the room map.	V2	S
FR-ADV-002	**Option A.** It shall optimize the racing line (minimum curvature, then minimum lap time) within the car's limits, with TUM's `global_racetrajectory_optimization` as the reference, and compute the speed profile along it.	V2	S
FR-ADV-003	**Option A.** It shall track the line at speed with a controller that copes with high speed (tuned Pure Pursuit, or MPC).	V2	S
FR-ADV-011	**Option B, hybrid localization.** The stack shall fuse wheel odometry, IMU and LiDAR localization in the room map with an EKF (for example `robot_localization`).	V2	S
FR-ADV-012	**Option B.** The detected tape lines shall correct the pose relative to the track, where the LiDAR is weak.	V2	S
FR-ADV-013	**Option B.** When one sensor drops out or gives a wrong reading, the pose shall stay usable.	V2	S

Off-track report (OFF)

ID	Requirement	Phase	Src
FR-OFF-001	Optionally, a node may compare the car's footprint at the stack's pose with the tape lines and report when it believes the car is off the track. It only reports, never stops the car. **[TBD: OI-11]**	**[TBD]**	S

2.4 Non-Functional Requirements

2.4.1 Portability and variability

ID	Requirement	Phase	Src
NFR-VAR-001	The stack shall not depend on the track layout: any closed loop drawn as two boundary lines.	V1	S
NFR-VAR-002	The stack shall not depend on the room, which is mapped on site.	V1	S
NFR-VAR-003	Line detection should not depend on the floor or the tape colour, once those are known **[TBD]**.	V1	S
NFR-POR-001	The same stack, images and launch shall run on the simulator (Mac, PC) and on the car (Orin Nano).	V1	S

2.4.2 Performance

ID	Requirement	Target	Justification
NFR-PER-001	Top speed on the track	**[TBD]**	Not set by the scope. The sensor choice was made for 5–8 m/s (H).
NFR-PER-002	Scan rate	40 Hz	At 8 m/s a 10 Hz LiDAR leaves 80 cm between scans, too much for scan matching (H).
NFR-PER-003	Camera frame rate	$\geq$ 60 fps	At 8 m/s, 30 fps means 27 cm travelled per frame (H).
NFR-PER-004	Timestamp error	$\leq$ 10 ms	10 ms at 8 m/s is 8 cm on the floor (H).
NFR-PER-005	Command timeout	0.2 s per mux input	Today's `ackermann_mux` configuration (C).

2.4.3 Safety

- **NFR-SAF-001:** a person can always stop or take over the car, on the car and in the simulator (FR-SAF-001 to 005, FR-SIM-002).
- **NFR-SAF-002:** each safety feature is tested on the car (AC-V1-04) before autonomous laps on the car.

2.4.4 Maintainability

- **NFR-MNT-001:** the forks stay upstream plus our upstreamable fixes; our code lives in `polimi_*` packages in our repo.
- **NFR-MNT-002:** pinned dependencies are bumped on purpose, never by following a branch.

2.5 Summer school exercises (reference)

The exercises go from the simplest to the most advanced; each builds on the ones before. The scope says each must run in the simulator first, then on the car. Whether that makes them requirements is open **[TBD: OI-12]**: the suggestion is reference only, with the simulator covering what the base platform's acceptance needs.

Exercise	Goal	Platform part
1. Trajectory tracking control (controllo e inseguimento traiettoria)	Follow a given path with a feedback controller. 1a with absolute localization (map path), 1b with computer vision (camera path). Same controller, only the reference changes.	FR-PTH
2. Line recognition (riconoscimento delle linee)	Detect the tape lines in the image and give their position on the floor, relative to the car. Input of 1b.	FR-LIN
3. Localization on a map (AMCL)	Map the room with SLAM, then localize in it with a particle filter. Input of 1a.	FR-MAP, FR-LOC
4. Planning (pianificazione)	Compute the path and speed profile, introduction to the racing line.	FR-ADV (A)
5. Hybrid localization (localizzazione ibrida)	Fuse odometry, IMU, LiDAR and camera with an EKF.	FR-LOC-003, FR-ADV (B)

3 System Design

3.1 Architecture Overview

The design is in `TDD_F1tenth`. In short: the stack has three layers. The logic (SLAM, localization, line detection, path following) and a common layer (command mux and safety) are the same on car and simulator. Below them, exactly one hardware layer serves a fixed set of ROS topics and TF frames: the car's drivers, or the simulator. The stack talks to the hardware only through that set.

3.2 System Components

Component	Requirements
Drivers	FR-DRV
Mapping, localization	FR-MAP, FR-LOC
Line detection	FR-LIN
Path following	FR-PTH
Mux and safety	FR-SAF
Simulator	FR-SIM
Advanced topic	FR-ADV

4 Implementation Guidelines

4.1 Technology Stack

ROS 2 Jazzy, `rmw_zenoh_cpp`, Docker images per machine type (`arm` Apple Silicon, `x86` PC, `jet` Orin Nano), `f1tenth_system` drivers, `orbbec_camera`, `slam_toolbox`, `nav2_amcl`, `robot_localization`. Details in `TDD_F1tenth`.

4.2 Development Standards

Every change goes through a pull request; the shared branches are protected. Python through `uv`. Runbooks in `docs/software/`.

4.3 Testing Strategy

Every function runs in the simulator first, then on the car. The acceptance below is checked on both.

4.3.1 Acceptance criteria

ID	Done when	Covers
AC-V1-01	The car maps the room and localizes with AMCL.	FR-MAP, FR-LOC
AC-V1-02	With the PID it drives full laps of the taped track without leaving the lines, on the track centre seen by the camera, on the car and in the simulator.	FR-LIN, FR-PTH, FR-SIM

ID	Done when	Covers
AC-V1-03	The same, on a path drawn in the map.	FR-PTH-008, FR-LOC
AC-V1-04	Tested on the car: pressing the emergency stop, releasing the deadman, taking over with the joystick and an obstacle in front each stop the car or hand it to the person.	FR-SAF
AC-V2-A	Option A: the car laps faster on the optimized line than on the centre line without leaving the lines, measured as lap times and line crossings over several laps.	FR-ADV-001 to 003
AC-V2-B	Option B: the fused pose is more accurate and more robust than LiDAR-only localization, measured against a ground truth (the simulator, or marks on the floor) at increasing speeds.	FR-ADV-011 to 013

Leaving the lines is judged by the people watching: nothing in software measures it on the car, and there is no external tracking (C-005, FR-OFF-001).

5 Project Timeline

Phase	Start	End
Hardware selection	Sep 2026	7 Oct 2026 (BOM)
Simulator on Humble	Oct 2026	done
Jazzy migration (OI-13)	**[TBD]**	**[TBD]**
Base platform (V1)	**[TBD]**	**[TBD]**
Advanced topic (V2)	after V1 runs	**[TBD]**
Summer school	**[TBD]**	**[TBD]**

6 Risks and Mitigation

Risk	Prob.	Impact	Mitigation
Chassis is end of life, low stock	Medium	High	Several retailers in the BOM; Plan B parts list in `TDD_F1tenth`, Hardware Selection (H)
Glare or floor reflections hide the tape	Medium	High	Early phone video of the real tape; method choice OI-06; global-shutter camera
Symmetric rectangular room confuses localization	Medium	Medium	Always give the initial pose (OI-07)
People or moved furniture change the room after mapping	High	Medium	Re-map on site (A-001); option B corrects from the tape
Camera pitch under braking breaks the floor projection	Medium	Medium	Floor plane from depth, or the IMU (H)

Risk	Prob.	Impact	Mitigation
No USB passthrough on Docker Desktop for macOS: no joystick in the simulator	High	Medium	A stand-in for the joystick in the simulator **[TBD]**
A gain from option A too small to measure in a small room	Medium	Medium	Decide OI-01 after V1 runs

7 Open Items

Decisions and checks the requirements leave open, folded from the scope with their options and suggested answers. Nothing is decided until the row says so. [DECIDE] is a choice for the team, [VERIFY] a fact to check.

ID	Item	Options and suggestion	Affects
OI-01	[DECIDE] Advanced topic: option A (racing line) or B (hybrid localization). **Held** until the base platform runs (21 Oct 2026).	Both need the tape lines in map coordinates (OI-03). A also needs the vehicle parameters (friction, CoG height), the TUM optimizer, a controller that holds at speed and a localization that keeps up; easy to prove (lap times, line crossings), but the gain is small in a small room. B needs an EKF, a camera correction that observes only the lateral offset and heading to the tape, and handling of sensor failures; harder to prove on the car (needs a ground truth) and overlaps exercise 5.	FR-ADV
OI-02	[DECIDE] Frame of the line-detection path: `base_link` or `odom`.	Suggested: `odom`, transformed at the image time stamp. It compensates the camera latency, keeps a fragment valid between frames, and turns the stale-path rule into "keep going on the last path for a short distance, then stop". Alternative: `base_link`, with the PID looking up the transform at the image time. The path-following contract (any frame) does not change.	FR-LIN-002
OI-03	[DECIDE] Where the path drawn in the map comes from. The tape is not in the LiDAR map, so the path cannot be drawn by hand inside it.	Suggested for the base platform: teach and repeat: drive the camera follower (1b) slowly with AMCL running, record the poses in the map, smooth them, save them as the path. Later, for A or B: tape mapping, projecting the detected tape into the map to build the two boundary lines.	FR-PTH-008
OI-04	[DECIDE] Open paths. Exercise 1a starts from a straight line, but offline paths are published as closed loops.	Add an open-path mode: the PID stops the car at the end of the path.	FR-PTH-005
OI-05	[DECIDE] Form of the PID. A PID on the lateral error alone oscillates as speed rises.	Lateral and heading error with a look-ahead point (Stanley-like); Pure Pursuit as the drop-in for higher speed.	FR-PTH-001

ID	Item	Options and suggestion	Affects
OI-06	[DECIDE] Line detection method.	(a) Image space: colour threshold, bird's-eye view, sliding windows, polynomial fit (the classic exercise 2). (b) Floor points: segment the tape, project the pixels onto the floor, fit and track the lines in `odom` (also feeds tape mapping, OI-03). (c) A learned segmentation, only if thresholds fail on the real floor. Hard cases for all: hairpins with three or four lines in view, one line in view, glare. First data: phone video of the real tape.	FR-LIN-001
OI-07	[DECIDE] Localization node.	`nav2_amcl` (named by the scope; in the image with DEC-02 of the TDD), `slam_toolbox` in localization mode, or MIT `particle_filter` if AMCL is too slow at speed. Risks: a symmetric rectangular room (always give the initial pose), people around the track, furniture moved after mapping.	FR-LOC-001
OI-08	[DECIDE] EKF for `odom` (wheel odometry and the camera IMU, `robot_localization`) in the base platform.	Suggested: yes. It helps exercise 1a and both options, and is the first half of exercise 5.	FR-LOC-003
OI-09	[VERIFY] Safety with `ackermann_mux` locks.	Emergency stop: a latched lock above the joystick. Autonomy deadman: a lock with a timeout between `/drive` and the joystick, released while RB is held (today RB's `autonomous_control` publishes to `/dev/null`, so the autonomy is not gated). AEB: an input between `/drive` and the joystick that brakes under a time-to-collision threshold, so a person can still drive the car away. Check how locks behave in this mux.	FR-SAF-002, 003
OI-11	[DECIDE] Off-track report from the stack's own estimate (optional). On the car, leaving the lines is judged by the people watching; there is no external tracking.	A node that compares the car's footprint at the stack's pose (map or odom) with the tape lines (tape map or detected lines) and reports when it believes the car is off the track. It only reports, never stops the car, and shows what the stack believes, not the truth.	FR-OFF-001
OI-12	[DECIDE] Exercises in the simulator: reference or required.	Suggested (user, 21 Oct 2026): reference only, not required in the simulator or the base platform; the simulator covers what the base platform's acceptance needs. Optional simulator additions for the exercises are in simulation plan 01.5 (O-25).	section 2.5

Decided: OI-10 (one image rebuild with `nav2_amcl`, `robot_localization`, `image_transport_plugins`, `cv_bridge`; 22 Oct 2026) and OI-13 (Jazzy, C-020; 21 Oct 2026).

8 Glossary and References

8.1 Definitions

30L 30 e lode, the top grade.
AEB Automatic emergency braking.
AMCL Adaptive Monte Carlo localization, a particle filter on a known map.
Deadman A button the operator holds for the autonomy to drive; releasing it stops the car.

EKF Extended Kalman filter.

Mux The command multiplexer (`ackermann_mux`): forwards the highest-priority input still sending.

SLAM Simultaneous localization and mapping.

VESC The motor controller; it also drives the steering servo and gives wheel odometry.

8.2 References

1. Project scope, 21 Oct 2026: `docs/scope.md` at commit `425ee04`, folded into this document and removed.
2. Hardware studies (LiDAR 23 Sep, chassis 24 Sep, camera 7 Oct 2026): folded into `TDD_F1tenth`, Hardware Selection.
3. `hardware/bom/master_bom.xlsx`, updated 7 Oct 2026.
4. `TDD_F1tenth`, the design that refines these requirements.